Master AAI Expert Workshops — Session 1

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Agenda

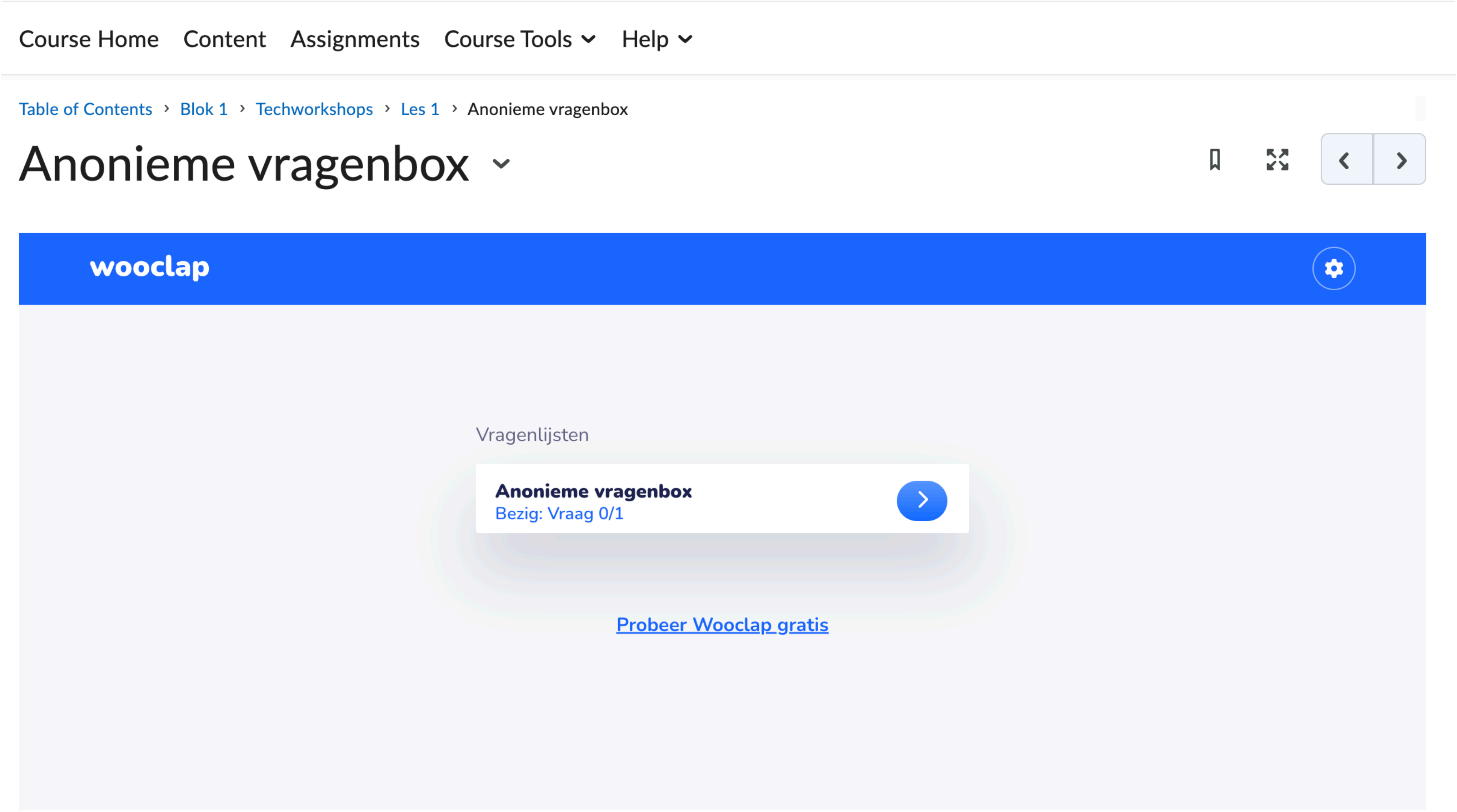
(SmartArt diagram — see original slides)

Introduction Tech Workshop Curriculum

Curriculum Overview

No.	Date	Lecturer	Subjects
1	Thu Sep	Rick + Michiel	AI assisted coding, Data exploration, Faces
2	Thu Sep	Michiel	Object Detection, Video, CNN advanced
3	Thu Sep	Rick	Evaluations, Build an app
4	Thu Sep	Michiel	Embeddings (SSL, CLIP, Search) & prep tech reviews
5	Thu Oct	Rick + Michiel	Tech reviews
6	Thu Oct	Rick	ML Ops + topics based on student questions
7	Thu Oct	Michiel	Vision Language Models (VLMs) + topics based on student questions

Vragenbox op DLO



Techreview

Assessment on the technical aspect of your project up till that point. Outcomes A2, B1, B2, B3, C2

You hand in on the DLO:

- Relevant part of your project report (see study guide)
- Link to your up-to-date Git repo (make it navigable)

See it as a feedback moment on your progress, not as a test.

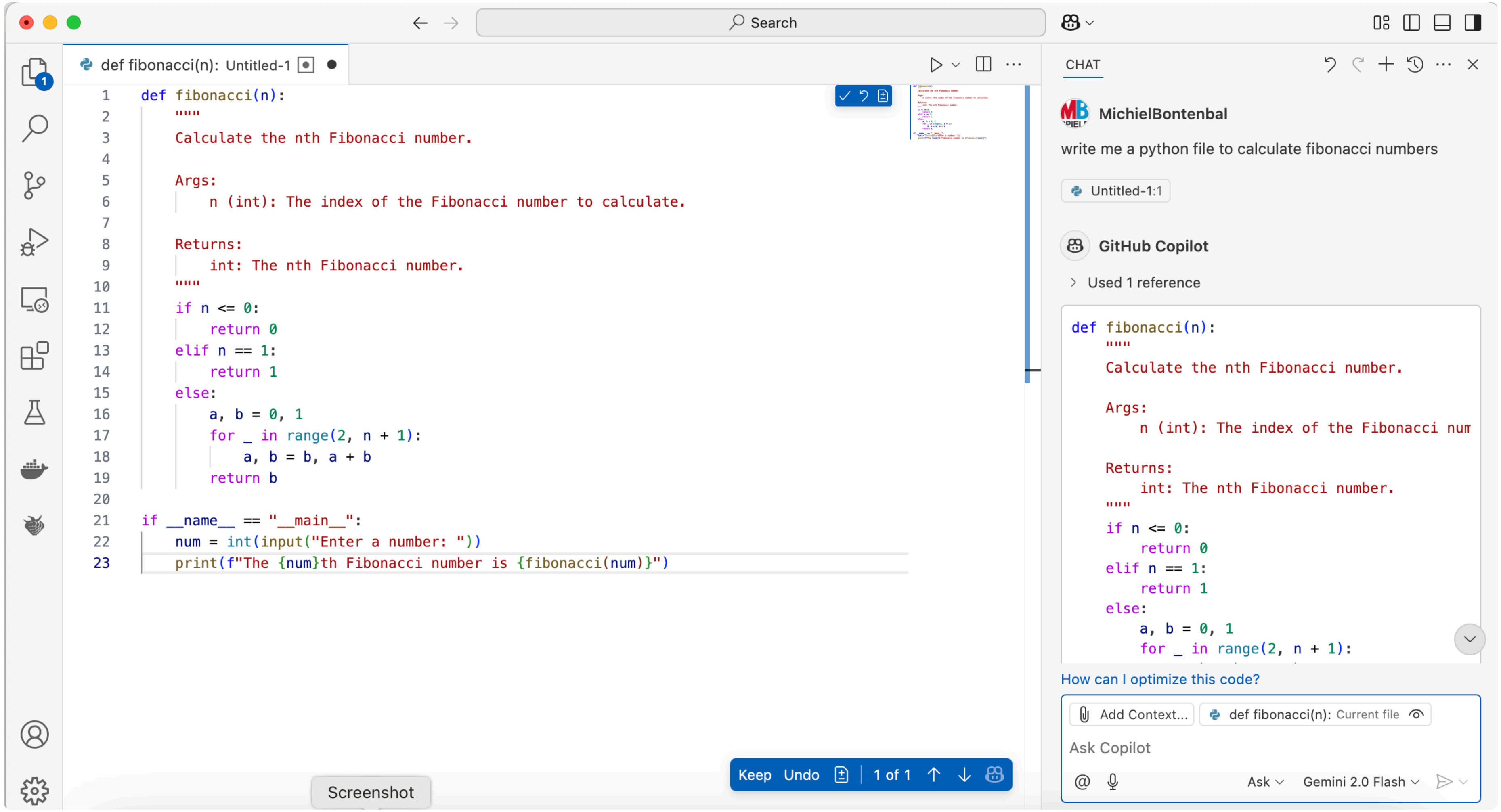
Usage of Generative AI

AI Assessment Scale Level 3: AI mag gebruikt worden ter ondersteuning bij specifieke taken. Je wordt altijd verplicht om AI-gegenereerde content kritisch te evalueren en indien nodig aan te passen (Zie Techworkshop DLO).

Verantwoording van keuzes in eigen woorden om begrip aan te kunnen tonen — Gen AI niet toegestaan voor schrijven uiteindelijke uitleg in rapport.

AI Assisted Coding

AI Assisted Coding in VS Code



Try a different model!

Do you get the right response? ‘Alignment’

Intro to Alignment: aligned.substack.com — blog by Jan Leike

Tips for Better Prompting

Write a better prompt

- Think before you prompt
- Try different prompts
- For beginners: add “write simple code…”

Understand the response

- Be aware of mistakes
- Ask “Explain this code to me”
- Decide if you want to use the code or try again

Human–AI Pair Programming

Work together with the AI — your new best buddy!

Active Learning is Better!

Writing code is more effective than reading code. Why?

- You remember it better
- You are more critical
- You are more engaged

First Learn to Walk Before You Learn to Run

To Do / Exercises

Beginners

- Set up Copilot in VS Code
- Write code to calculate Fibonacci numbers

Experienced

- Compare: Claude.ai vs Copilot
- Try uploading an image to generate code
- Create a front-end app with HTML/JS/CSS (multiple files)

Data Exploration and Initial Analysis

The Life-Cycle of MLOps: CRISP-ML(Q)

<table>
<tr><th>CRISP-ML(Q)</th><th>CRISP-DM</th><th>Amershi et al. [28]</th><th colspan="2">Breck et al. [29]</th></tr>
<tr><td rowspan="2">Business & Data Understanding</td><td>Business Understanding</td><td>Requirements</td><td colspan="2">-</td></tr>
<tr><td>Data Understanding</td><td>Collection</td><td rowspan="4">Data</td><td rowspan="7">Infrastructure</td></tr>
<tr><td rowspan="3">Data Preparation</td><td rowspan="3">Data Preparation</td><td>Cleaning</td></tr>
<tr><td>Labeling</td></tr>
<tr><td>Feature Engineering</td></tr>
<tr><td>Modeling</td><td>Modeling</td><td>Training</td><td>Model</td></tr>
<tr><td>Evaluation</td><td>Evaluation</td><td>Evaluation</td><td>-</td></tr>
<tr><td>Deployment</td><td>Deployment</td><td>Deployment</td><td>-</td></tr>
<tr><td>Monitoring & Maintenance</td><td>-</td><td>Monitoring</td><td colspan="2">Monitoring</td></tr>
</table>

Choosing a Dataset

- Always first concern yourself with **Data Provenance**
  - Source, collection process — might any of these be biased?
- Then check **Data Description**
  - Size, contents, label quality
- **Where to find data?**
  - Similar research, Kaggle, Hugging Face

Initial Exploration and Visualisation

- Visualisation is an integral part of the data exploration phase
- First look at different distributions: feature distributions, bivariate/multivariate exploration
- Use histograms, boxplots, scatterplots, etc.

Also check technical details:

- Target leaking
- Image quality (resolution, brightness, cropping)
- Label noise (incorrect ages, approximations)

Ask yourself: what makes this data predictive?

Data Bias When Using People

- Algorithms do not care about your data at all
- They just want to “win” by solving your problem
- Winning = minimising your chosen loss function
- By finding any pattern that might contribute to this
- **Representation is therefore unimaginably important**

Feature Engineering

When you know your data, you can start engineering:

- Face alignment / Cropping
- Grayscaling
- Over- and Undersampling of classes
- Normalisation of the data
- Regression vs Classification framing

Don’t just ask yourself what — but more importantly why.

Other Important Subjects

- Data leakage / test data invalidation
- Data splitting strategies (random, stratified, etc.)
- Think already about evaluation metrics (MAE, RMSE, Fairness metrics)
- Reproducibility of your workflow
- The iterative nature of it (learning outcome B4)

Practicum ~45 min

Make groups of four students. Perform data analysis on the **UTKFace dataset** (see notebook on DLO).

Questions to answer:

- **Distribution:** What distributions should be checked? Balanced or skewed?
- **Quality:** Any errors, anomalies, missing data, unrealistic images?
- **Bias:** What kind of biases are present? How would you fix them?
- **Image inspection:** Lighting, angles, cropping, etc.

In the end we recap: surprises? concerns? take-aways for the project?

Faces

Faces — Time Lapse

Python Packages for Faces

No	Name	Description	Link
1	MTCNN	CNN model for face detection	pypi.org/project/mtcnn
2	Deepface	CNN model for age estimation	pypi.org/project/deepface
3	face-recognition	Face recognition library	pypi.org/project/face-recognition

Exercise Notebooks

- `Gezichtsdetectie_with_cnn.ipynb`
- `Age_estimation_Deepface.ipynb`